Walmart Sales Forecast EDA

Jericho Jacala

Walmart Sales Forecast / Exploratory Data Analysis

Summary:

The purpose of this document is to report the findings of exploratory data analysis (EDA). Through direct analysis of historical sales data for 45 Walmart stores from 2010-2012, we intend to identify the key characteristics of the data, preliminary business strategy, and design a machine learning model which can make sales forecasts.

Contents

1 FINDINGS

The purpose of this document is to report the findings of exploratory data analysis (EDA). This document ONLY documents EDA and is not the final report. Through direct analysis of historical sales data for 45 Walmart stores from 2010-2012, we intend to identify the key characteristics of the data, preliminary business strategy, and design a machine learning model which can make sales forecasts.

All pertinent documents are maintained in a github repository.

1.1 DATASET DESCRIPTION

The dataset, which was found on Kaggle, contains historical data from the supermarket chain Walmart. It was the most in-depth free sales data we found and its weekly and store-by-store data points allow us to take advantage of examining the performance of each store and how seasonal events affect sales.

Features of the dataset include the following:

- Store - the store number
- Date - the week of sales
- Weekly_Sales - sales for the given store
- Holiday_Flag - 1 for special holiday weeks, 0 otherwise
- Temperature - temperature on the day of that week
- Fuel_Price - cost of fuel in the region
- CPI - prevailing consumer price index
- Unemployment - prevailing unemployment rate

Some of the descriptions of the features from the Kaggle website remain quite vague. For example, since all dates occur on Fridays, we are not sure what day of the week sales data begins and ends. We assume that each sales cycle ends on Friday, meaning that they begin on the

Saturday before that Friday. Additionally, assumptions were made that temperature is measured in Fahrenheit, gas prices are measured in USD per gallon, and CPI is measured using 1984 as a base year. It is important to note that the Super Bowl, Labor Day, Thanksgiving, and New Year's are the special holiday weeks that Walmart prioritizes using special markdown events. Notably, the week of Christmas is left out as a holiday despite higher sales on this week than any other week.

1.2 DATA ENGINEERING

Various transformations were made to the data through Power Query in PowerBI. The dates for each data point were initially in text and DD-MM-YYYY format. The dates were changed to a date and MM-DD-YYYY format that PowerBI more easily recognizes. Two formatted holiday features were added: FormattedHolidayFlag is "Holiday" for each record that is a holiday and "Not Holiday" for all other days. HolidayType records the type of holiday (Christmas, Super Bowl, Labor day, New Year's, or Thanksgiving). With these features, we can discern between holidays and non-holidays and even between the different types of holidays. Based on the high number of sales the week of Christmas (12/24/2010 and 12/23/2011) we decided to include Christmas as a Holiday in the new features. One of the conclusions drawn from this EDA is that Christmas should be included as a holiday and new strategies should be used to prioritize it (see Data Analysis).

List of exact dates for each holiday:

- February 12, 2010 (Super Bowl Weekend)
- September 10, 2010 (Labor Day Weekend)
- November 26, 2010 (Black Friday/Thanksgiving)
- December 24, 2010 (Christmas Eve)
- December 31, 2010 (New Year's Eve)
- February 11, 2011 (Super Bowl Weekend)
- September 9, 2011 (Labor Day Weekend)
- November 25, 2011 (Black Friday/Thanksgiving)
- December 23, 2011 (Christmas Eve)
- December 31, 2011 (New Year's Eve)
- February 10, 2012 (Super Bowl Weekend)
- September 7, 2012 (Labor Day Weekend)

Please note that the Christmas Eve weeks were subsequently added to the data set in the cleaning phase.

1.3 DATA ANALYSIS

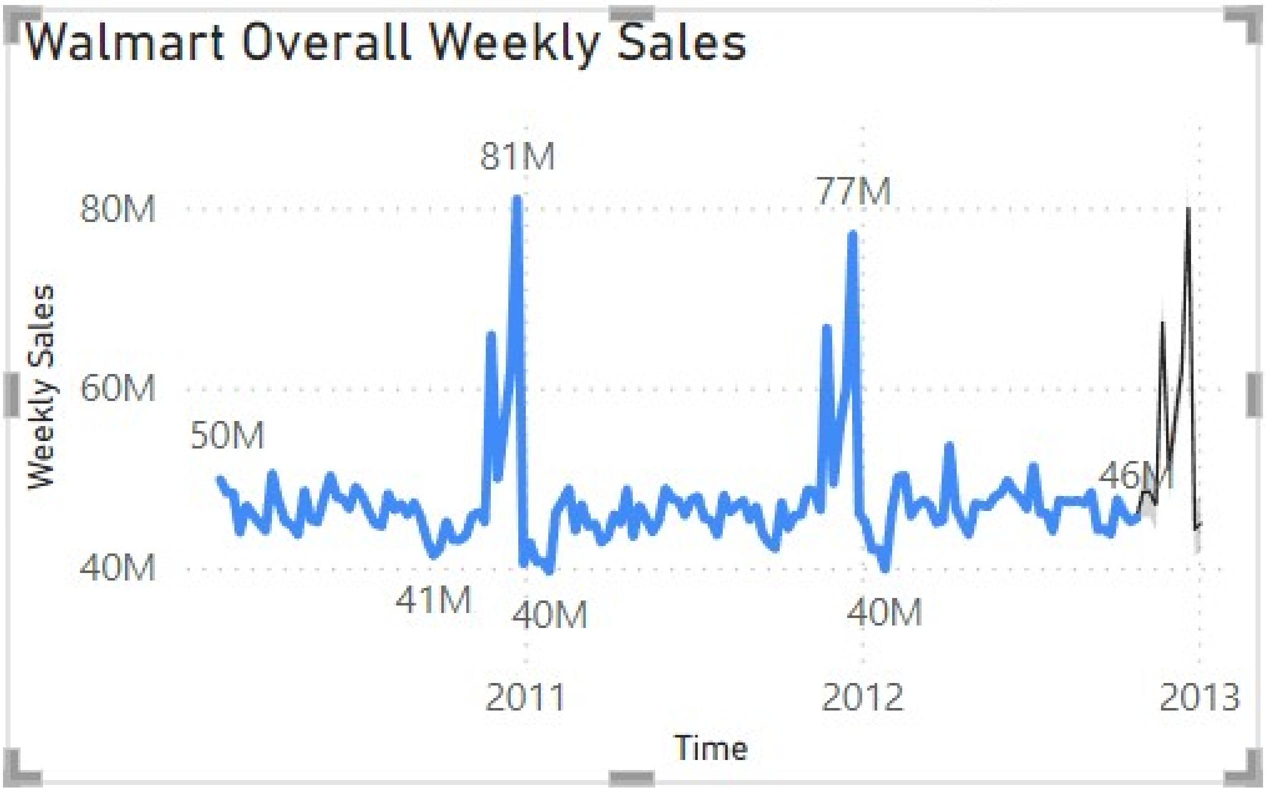
1.3.1 OVERVIEW

In this study we noted several key characteristics. Most notably, overall sales numbers seem to follow yearly cycles with spikes at certain holidays. For an interactive report, please visit the repository.

1.3.2 WALMART OVERALL WEEKLY SALES

By combining the sales from each store for each week, we compiled a line chart of overall Walmart sales over time. Using the forecast tool in Power BI analytics, we generated a projection of Walmart sales until the end of 2012, as seen in black. By setting seasonality to 52, we are able to get realistic forecasts for the future. Seasonality is the number of data points it takes to complete a "cycle." The data appears to follow a yearly cycle, so given the data is weekly and there are 52 weeks in a year, seasonality is set to 52. Notably, Thanksgiving 2012 is projected to have 67 million in sales, and Christmas is projected to have 80 million. There does not appear to be any obvious trends in the data. The data appears to be stationary.

The seasonal forecasting tool uses the Holt-Winters method, also known as Exponential Time Smoothing Additive Error, Additive Trend, Additive Seasonality, or ETS AAA for short. The algorithm minimizes mean squared error for the training window and determining a validation window where the state is not adjusted at each step but the MSE is calculated for the whole validation window. This technique improves trend preservation. [PowerBI(2014)].

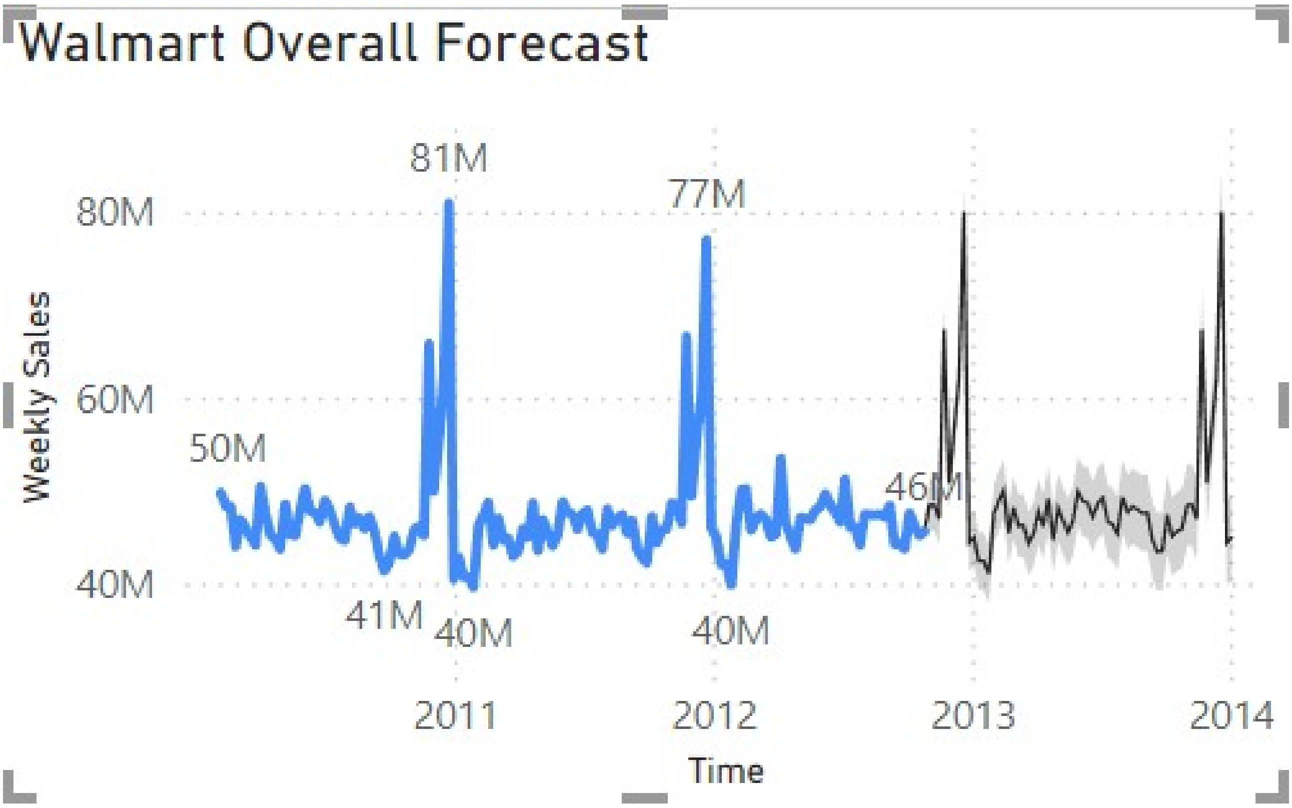


WALMART OVERALL WEEKLY SALES

The graph displays a clear cyclical pattern with spikes around the Thanksgiving and Christmas holidays. It is also important to note that the weeks with lowest sales appear to come during New Year's and the weeks following. Sales generally appear to hold constant over time with no significant trends from year to year.

1.3.3 WALMART OVERALL SALES FORECAST TO END OF 2013

Using the Holt-Winters algorithm, we developed a forecast through 2013. The gray area marks a 95% confidence interval. The data appears very similar to our historical data, with similar peaks around Thanksgiving and Christmas. For 2013, we can expect a total of 2.55 billion in sales. Under the Holt-Winters model, we are 95% confident that sales for 2013 will remain between 2.35 and 2.76 billion. This, of course, does not account for economic, strategical, or otherwise significant changes for 2013.



WALMART OVERALL FORECAST

1.3.4 AVERAGE OVERALL WEEKLY SALES BY HOLIDAY

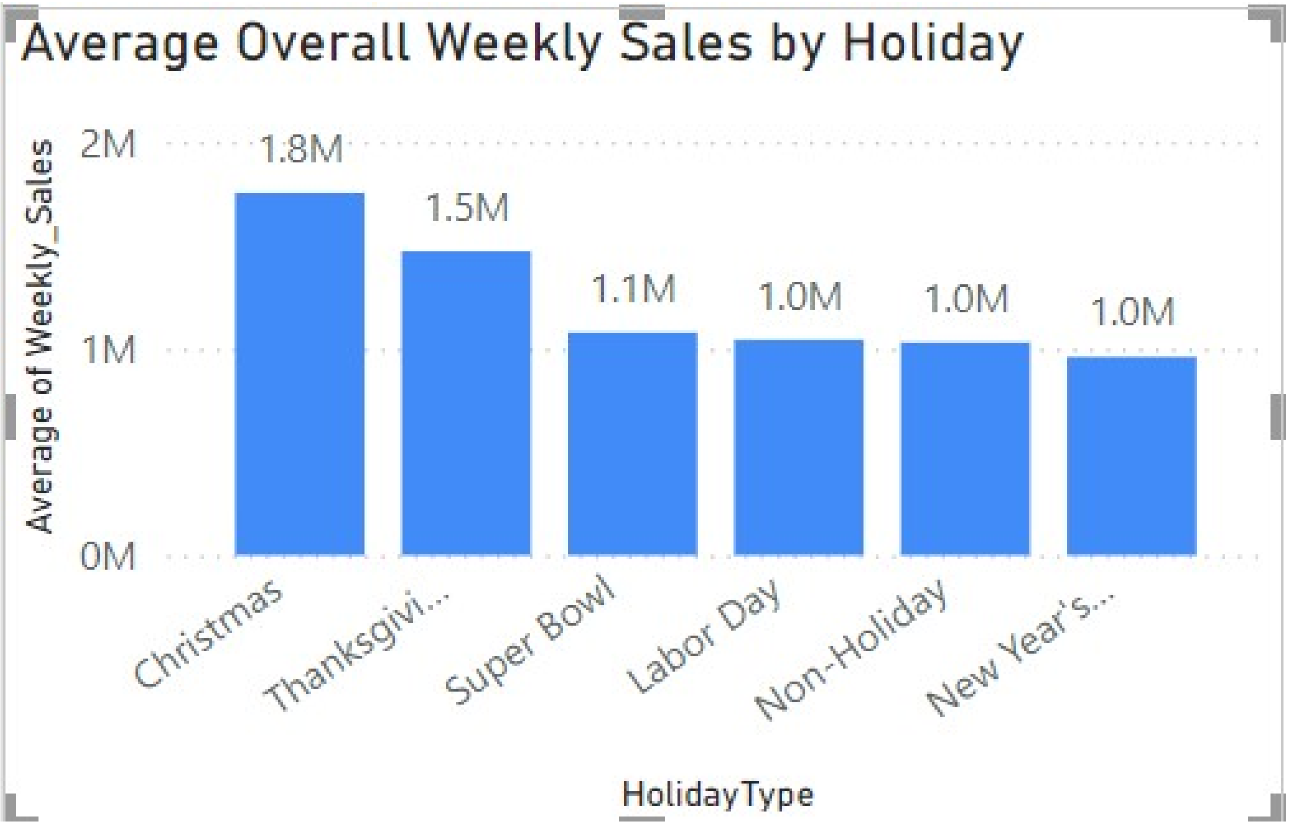
By determining how much sales each holiday brings in on average, we can determine whether there is a difference in sales between holidays and, if so, what holidays bring in the most sales.

Christmas and Thanksgiving appear to yield the most sales with 1.8 million and 1.5 million in sales respectively. This is much larger in contrast with the 1.1 million averages of the Super Bowl and the averages of Labor Day, non-holidays, and New Year's, which all average about 1 million in sales. The high number of sales on Christmas may suggest that Walmart should develop strategies to prioritize increasing sales on Christmas. According to the dataset description on Kaggle, promotional markdowns for holidays only occur on the Super Bowl, Labor Day, Thanksgiving, and New Year's. Consumer retail spending appears to be extremely high during Christmas time, meaning developing strategies to win more of this spending in sales could prove lucrative for Walmart.

2 MACHINE LEARNING MODEL DESIGN PROPOSAL

2.1 MODEL RESEARCH

Time series forecasting is a method of estimating future values based on historic trends and patterns [GeeksForGeeks(2023)]. Using time series forecasting is the most direct and effective



AVERAGE OVERALL WEEKLY SALES BY HOLIDAY

way to forecast future data. We will consider two different time series models: Autoregressive Integrated Moving Average (ARIMA) and Long Short-Term Memory (LSTM).

2.1.1 AUTOREGRESSIVE INTEGRATED MOVING AVERAGE

Autoregressive models forecast variables through a linear combination of past values of the variable. These models are usually restricted to stationary data. This can be thought of in contrast to multiple regression models, which use a linear combination of other features to forecast. We can write out an autoregressive model of order q as:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + ... + \phi_p y_{t-p} + \epsilon_t$$

The order represents the number of previous values to consider. ϵ_t represents white noise [Athanasopoulos and Hyndman(2018)]. To find the best fit, we must find the set of ϕ values that minimize $\sum_{t=1}^{T} \epsilon_t^2$, where T is the number of observations.

Moving average models use past forecast errors rather than past values to forecast [Athanasopoulos and Hyndman(2018)]:

$$y_t = c + \epsilon_t + \phi_1 \epsilon_{t-1} + \phi_2 \epsilon_{t-2} + ... + \phi_q \epsilon_{t-q}$$

References

[Athanasopoulos and Hyndman(2018)] George Athanasopoulos and Rob J. Hyndman. *Forecasting: Principles and Practice*. OTexts, second edition, 2018.

[GeeksForGeeks(2023)] GeeksForGeeks. Time series forecasting using pytorch, 2023. URL `https://www.geeksforgeeks.org/time-series-forecasting-using-pytorch/`.

[PowerBI(2014)] PowerBI. Describing the forecasting models in power view, 2014. URL `https://powerbi.microsoft.com/fr-ch/blog/describing-the-forecasting-models-in-power-view/`.